

Problem Set for 12-May and 15-May

The ambient field is $\mathbb{R}$.

Exercise 1 Determine whether the following are inner product spaces (简要回答即可).

1. V is the vector space continuous functions over $\mathbb{R}$, and the pairing is

$$(f, g) \mapsto \int_0^1 f(x)g(x) \, dx.$$

2. $V = \mathbb{R}[x]$, and the pairing is

$$(f, g) \mapsto \int_{-\infty}^{\infty} f(x)g(x)e^{-x^2} \, dx.$$

3. $V = \mathbb{R}^{n \times n}$, and the pairing is

$$(A, B) \mapsto \det(A^T \cdot B).$$

4. (**optional**) By axiom of choice, any $\mathbb{R}$ -vector space can be endowed with an inner product structure.

Exercise 2 Show that every real matrix is similar to a block diagonal matrix, wherein the candidates of the blocks are

- the Jordan form of a real eigenvalue;
- the block of the form

$$\begin{bmatrix} H_{r,\theta} & Q & & & \\ & H_{r,\theta} & Q & & \\ & & \ddots & \ddots & \\ & & & H_{r,\theta} & Q \\ & & & & H_{r,\theta} \end{bmatrix},$$

where

- $H_{r,\theta} = \begin{bmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{bmatrix}$, and
- $Q = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

Note

When proving with Jordan form, show that $A \sim B$ over $\mathbb{R}$ if and only if $A \sim B$ over $\mathbb{C}$.

Exercise 2.5 Show that every real matrix is a product of two real symmetric matrices.

Exercise 3 Let $(V, (-, -))$ be a finite dimensional real inner product space. For unit vector v_0 (i.e., $(v_0, v_0) = 1$), we define the reflection

$$\varphi : V \rightarrow V, \quad x \mapsto x - 2(x, v_0) \cdot v_0.$$

Show that for isometric transform ψ (i.e., $(\psi(u), \psi(v)) = (u, v)$), 1 is an eigenvalue for ψ or $\psi \circ \varphi$. Overall, $1 \in \sigma(\psi) \cup \sigma(\psi \circ \varphi)$.

Note

In short, if 1 is not the eigenvalue of ψ , then

- 1 must be an eigenvalue with the composition of any reflections.

Exercise 4 Set $V := \mathbb{R}[x]$ and $V_0 := \{f \in \mathbb{R}[x] \mid f(0) = f(1)\}$.

1. Prove that $V \times V \rightarrow \mathbb{R}, \quad (f, g) \mapsto \int_0^1 f(x)g(x) \, dx$ is an inner product.
2. Set $\mathcal{D} : V_0 \rightarrow V, \quad f(x) \mapsto f'(x)$. Find $\dim \ker(\mathcal{D})$ and $\dim \operatorname{coker}(\mathcal{D})$.
3. Define the inner product restricted on the subspace

$$(\cdot, \cdot)_0 : V_0 \times V_0 \rightarrow \mathbb{R}, \quad (f, g) \mapsto \int_0^1 f(x)g(x) \, dx.$$

Is there any linear map $\mathcal{D}^* : V \rightarrow V_0$ such that for any $h \in V_0$ and $g \in V$,

$$(\mathcal{D}^*g, h)_0 = (g, \mathcal{D}h)?$$

Tip

去年期末考.

Exercise 5 Let $(V, (,))$ be an inner product space, where V is not necessary finite dimensional.

- Say φ^* is the adjoint of $\varphi \in \operatorname{Hom}_{\mathbb{R}}(V, V)$, provided $(\varphi^*(x), y) = (x, \varphi(y))$ for arbitrary $x, y \in V$.
- Let U be a subspace of V . Set

$$U^\perp := \{v \in V \mid (u, v) = 0, \forall u \in U\} = \bigcap_{u \in U} \ker((u, -))$$

Now we assume that φ is invertible, and φ^* exists.

1. Show that φ^* is injective, and $(\operatorname{im}(\varphi^*))^\perp = 0$.
2. Show that φ^* is surjective $\implies (\varphi^{-1})^*$ exists and $(\varphi^{-1})^* = (\varphi^*)^{-1}$.
3. Show that $(\varphi^{-1})^*$ exists $\implies \varphi^*$ is invertible and $(\varphi^{-1})^* = (\varphi^*)^{-1}$.

4. Let $V = \mathbb{R}[x]$ and $(f, g) := \sum_{n \geq 0} f_i g_i$, where $h = \sum h_i \cdot x^i$. Show that
1. $(V, (,))$ is an inner product space;
 2. $L : f \mapsto \frac{f-f(0)}{x}$ is the linear map of moving left. Show that $(\text{id} + L)$ is invertible;
 3. Show that $\varphi := (\text{id} + L)^{-1}$ has no adjoint φ^* .

Exercise 6 (Optional, very interesting). Find the dimension and inertia of the following symmetric bilinear form, and verify the invariant group action.

1. $V = \{X \in \mathbb{R}^{2 \times 2} \mid \text{tr}(X) = 0\}$, and the pairing is $(A, B) \mapsto \text{tr}(A \cdot B)$. Moreover, for $g \in \text{GL}_2(\mathbb{R})$, the pairing is invariant under $X \mapsto g \cdot X \cdot g^{-1}$.

Step 0 $\dim V = 3$.

Step 1 The pairing is linear (trivial) and symmetric, since $\text{tr}(AB) = \text{tr}(BA)$.

Step 2 Let $(A_1, A_2, A_3) := (E_{1,2}, E_{2,1}, E_{1,1} - E_{2,2})$ be a basis of V , the associated Gram matrix is the 3×3 symmetric matrix $(\text{tr}(A_i A_j))_{3 \times 3}$ is $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. The inertia is $(++-)$.

Step 3 Since $\text{tr}((gXg^{-1}) \cdot (gYg^{-1})) = \text{tr}(XY)$, pairing is invariant under such group action.

2. $V = \mathbb{R}^{2 \times 2}$, and the pairing is $(A, B) \mapsto \text{tr}(\text{adj}(A) \cdot B)$. Moreover, for $(g, h) \in \text{SL}_2(\mathbb{R}) \times \text{SL}_2(\mathbb{R})$, the pairing is invariant under $X \mapsto g \cdot X \cdot h^{-1}$.
3. $V = \left\{ \begin{pmatrix} a & ib \\ ic & \bar{a} \end{pmatrix} \mid a \in \mathbb{C}; b, c \in \mathbb{R} \right\}$, the pairing is $(X, Y) \mapsto \text{Re}(\text{tr}(X \cdot \bar{Y}))$. Moreover, for $g \in \text{GL}_2(\mathbb{C})$, the pairing is invariant under $X \mapsto g \cdot X \cdot (\bar{g})^{-1}$.
4. (**Optional**, if you know tensor product). Let $U \wedge U$ be the quotient space of $U \otimes U$ by the relation $(a \otimes b + b \otimes a = 0)$. In particular, $\dim(U \wedge U) = \binom{\dim U}{2}$. Let $V = \mathbb{R}^4 \wedge \mathbb{R}^4$ be the 6-dimensional vector space with basis $\{e_i \wedge e_j\}_{1 \leq i < j \leq 4}$, and the pairing is $(\lambda, \mu) \mapsto \lambda \wedge \mu$. Here $\bigwedge_{i=1}^4 \mathbb{R}^4$ is one-dimensional with basis $\{e_1 \wedge e_2 \wedge e_3 \wedge e_4\}$. Moreover, for $g \in \text{SL}_4(\mathbb{R})$, the pairing is invariant under $\mu \mapsto g \cdot \mu$. Here $g \cdot (\sum c_{i,j} e_i \wedge e_j) := \sum c_{i,j} (g \cdot e_i) \wedge (g \cdot e_j)$.

The above calculation provides the elementary construction of low dimensional spin groups, that is, the double covering of special orthogonal groups. One learns from these construction that

1. $\text{SL}_2(\mathbb{R})/\{\pm I\} \simeq \text{SO}(2, 1)$, and $\text{SL}_2(\mathbb{C})/\{\pm I\} \simeq \text{SO}_3(\mathbb{C})$;
2. $(\text{SL}_2(\mathbb{R}))^2/\{\pm(I, I)\} \simeq \text{SO}(2, 2)$, and $(\text{SL}_2(\mathbb{C}))^2/\{\pm(I, I)\} \simeq \text{SO}_4(\mathbb{C})$;

$$3. \mathrm{SL}_2(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}(3, 1);$$

$$4. \mathrm{SL}_4(\mathbb{C})/\{\pm I\} \simeq \mathrm{SO}_6(\mathbb{C}), \text{ and } \mathrm{SL}_4(\mathbb{R}) \simeq \mathrm{SO}(3, 3).$$

As corollaries, $\mathrm{SO}_4(\mathbb{C})$ is a double cover of $\mathrm{SO}_3(\mathbb{C})$, and the group isomorphism $\mathrm{SO}(3, 1) \simeq \mathrm{SO}_3(\mathbb{C})$ is a famous coincidence in quantum physics.